Meghana Zagabathuni

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Education

New York University — New York, NY
Master of Science in Computer Science — *Expected May 2027*
- **Coursework:** Algorithms, Software Engineering, Big Data, Machine Learning, Deep Learning

SRM Institute of Science and Technology — Chennai, India
Bachelor of Technology in Computer Science and Engineering; GPA: 8.96/10 — *2020 – 2024*

Projects

PawPal | *Python, Django, PostgreSQL, AWS Elastic Beanstalk, GitHub Actions*
- Owned the pets module of a **Django** pet-care marketplace on a 5-engineer Agile team, delivering 9 user stories
- Enforced per-user access so owners see only their pets and caregivers see care info only during active bookings
- Wrote 54 of the project's 248 tests, run in **GitHub Actions** CI with flake8 and a 90% coverage gate
- Added pet photo uploads with size validation and matched all pet pages to the team's **Figma** designs

CogniTron: AI Interview Coach | *Python, PyTorch, Hugging Face, Whisper, Llama 3.2*
- Created a voice AI coach that asks interview questions with **Llama 3.2** and transcribes answers with **Whisper**
- Fine-tuned 2 **DistilBERT** models with **Hugging Face**: a readability regressor and a 3-class difficulty classifier
- Achieved a 75.42 SUS score and a statistically significant gain in users' scores on a custom 0–100 depth metric

Dermatological Disease Detection and Skin Care Analysis | *Python, Streamlit, MySQL, CNN*
- Developed a web app that detects skin diseases, skin types and hair loss with a **CNN** at 92% accuracy

Experience

Teaching Assistant, Parallel Computing — Sep 2026 – Present
New York University — *New York, NY*
- Grade assignments and exams in an undergraduate course on multithreading and parallel performance analysis
- Enter grades and written feedback for each student on **Brightspace** with Teaching Assistant course access

Graduate Assistant, Finance Office — May 2026 – Aug 2026
New York University — *New York, NY*
- Cut report processing time by 30% by automating financial reporting and visualization in **Excel** and **Tableau**
- Reduced manual resolution effort by over 25% by automating repetitive processes across department web platforms
- Cut ticket turnaround by 10% by streamlining triage and prioritization of a high-volume **ServiceNow** queue

Core Team Member, Quantum Computing Team — Nov 2022 – May 2024
SRM Institute of Science and Technology — *Chennai, India*
- Researched quantum algorithm optimization with teammates and presented findings at meetings and conferences

AICTE AWS AI-ML Intern — Jan 2024 – Mar 2024
All India Council for Technical Education — *Remote*
- Deployed **TensorFlow** models on **AWS**, streamlining the deployment process and improving efficiency by 35%
- Completed hands-on **AWS** cloud and machine learning training through data-driven model projects

AI Intern — May 2022 – Jul 2022
Skill Vertex
- Trained a real-time Rock-Paper-Scissors hand-gesture recognizer in **PyTorch**, reaching 95% prediction accuracy
- Improved the model with real-time gesture data, then integrated and tested it with the team for live play

Publications

Z. Meghana et al. "Dermatological Disease Detection and Skin Care Analysis." *IRJMETS*, 2024
Z. Meghana et al. "Plant Leaf Disease Detection and Remedial System." *IJCRT*, 2024
Presented research at the F-COM'24 National Conference

Technical Skills

Languages: Python, C++, C, SQL, JavaScript, R, HTML/CSS, Shell scripting
Frameworks and Libraries: Django, FastAPI, REST APIs, PyTorch, TensorFlow, Hugging Face, Pandas, NumPy
Developer Tools: Git, GitHub Actions, AWS, CI/CD, Linux, PostgreSQL, MySQL, SQLite, Agile/Scrum, Unit Testing